Data Analytics 344
Memorandum

Section A

Question 1

(a) ZB is the largest
(b) Text, images, audio, video, social media

Question 2

Volume	The quantity of generated and stored data	Requires more memory and computational resources; requires data analytics tools to be scaled to large data sets
Variety	Data may be of different types, and from different data sources	Data analytics tools need to be adapted to work with different data types; Data transformations are required to get data into a format required by the data analytics tools
Velocity	The speed at which data is generated and processed	Requires data analytics tools to work in an online fashion; Data analytics tools should be computationally efficient; Require incremental models
Veracity	Refers to data quality	Data quality issues such as missing values, outliers, different scales, noise need to be corrected

Question 3

Value: By extracting hidden information/patterns/trends, value is added to data in the form of improved revenue, reduced costs, improved customer experience, etc.

Question 4

(a) Stationary refers to a fixed set of data; data set does not change over time.
Non-stationary refers to data streams, where data arrives over time.

(b) May result in concept drift, requiring the data model to be adapted over time; requires online analysis, or real-time analytics

Question 5

Data warehouses are central repositories of integrated data from different sources, storing enterprise-wide historical data.

Data marts store a subset of the data located in a data warehouse, oriented to a specific business line or team, or single department.

Data lakes are repositories of data stored in its natural raw format, usually as blobs.

Question 6

(a) Data analytics is the process of extracting useful insights from raw data; the process of discovering valuable information from very large databases using algorithms that discover hidden patterns in the data.
(b) Visual analytics utilises interactive visual interfaces to data for the exploration of data.
(c) Any of the following: unstructured data, similarity computation is difficult, similar images may look different, semantic queries on images are difficult without any meta data, noise, blurring, lighting, occlusions.

Question 7

(a) Predictive analytics
(b) Prescriptive analytics
(c) Diagnostic analytics

Section B

Question 1

Predictive data analytics is the art of building and using models that make predictions based on patterns extracted from historical data.

Question 2

(a) Yes, Model 1 is consistent because it correctly predicts each of the instances in the data set.

(b) Model 1 will generalise better to cases not in the data set, because the model is much simpler than model 2. Model 2 is too specific and overfits the data set.

Question 3

(a) The inductive bias of a learning algorithm:
    a. is a set of assumptions about what the true function we are trying to model looks like;
    b. defines the set of hypotheses that a learning algorithm considers when it is learning;
    c. refers to the error from incorrect assumptions in the learning algorithm
(b) If the bias is too high, then relevant relations between the features and targets can be missed, that is the model may underfit. In this case accuracy on both the training set and the test set will be bad. If bias is low, and variance is high, then the model may overfit.

Question 4

Supervised learning have as its objective to learn a functional mapping between a feature space and a target space. For each instance in the training set, a target value is provided. In supervised learning, no target features are provided, and the objective is to discover hidden patterns among the descriptive features.

Section C

Question 1

(a) Ordinal: ID, Policy Type
Nominal: Occupation, Pref Channel
Numeric: Age, Motor Value
Binary: Gender
(b) ID feature should be removed. It is nominal and has cardinality equal to the number of instances. Will have no correlation with the target features

Question 2

Removal of redundant, irrelevant features result in models that are less computationally expensive and memory intensive. Such features introduces noise, and increases chance of overfitting. Therefore, removal will result in less overfitting and better generalization.

Section D

Question 1

(a) The distribution in figure (b) is an exponential distribution. The large values in the tail indicate values that should be investigated, as they are likely to be outliers.

(b) The distribution in figure (d) is an uniform distribution, which may indicate unique values for the feature. If the cardinality of the feature is the same as the number of instances, then the feature should be removed.

Question 2

(a) If we set the number of bins to a very low number we may lose a lot of information about the original distribution. The fewer the number of bins, the more the resulting distribution becomes uniform. The more bins, the more similar the distribution of the original continuous feature is to the distribution of the binned representation. However, if f we set the number of bins to a very high number then we might have very few instances in each bin or even end up with empty bins. Binning also reduces noise.
(b) One-hot encoding should be used, as it maintains the order relation among the values of the original representation.

Question 3

(a) Three features have missing values:

Feature	**Motivation**	**Plan**	**Motivation**
H.R.Diff	13.03% from data quality table	Replace with mean over all available values	Numeric variable, with no outliers, and not too many missing values
PREV.TACHY	44.02% from data quality table	Possibly remove the feature	Too many missing values.
TACHYCARDIA	2.01% from data	Remove instances with	Cannot have instances

	quality table	missing values, so complete case removal; or impute by as most frequent class for nearest K neighbours	where target feature has missing values; for imputation, there are few few instances, so imputation based on similarity will not add a significant bias

(b) Three of the features stands out due to irregular cardinality:

Feature	**Motivation**	**Plan**	**Motivation**
AGE	Has a cardinality of 7 which is very low for a numeric feature	Rather convert to an ordinal feature	It may be better to rather refer to 7 age groups
GENDER	Has a cardinality of 4, but only 2 is expected	Standardise values by replacing male with m and female with f	There can be only two possible values for gender
TACHYCARDIA	Considering missing values, there are three possible values, while only two are expected	Nothing	The plan for missing values will remove this data quality issue

(c) Three features have problems with outliers

Feature	**Motivation**	**Plan**	**Motivation**
HEIGHT	Difference between the 1st quartile and the median values and the difference between the minimum and 1st quartile values are quite different; the histogram of the HEIGHT feature values shows significant skew to the left hand side	Remove the instances with the outlier values	Examining the data we see that there are a number of height values that are a couple of orders of magnitude less than the majority of values; such low heights not possible for provided ages
BMI	Unreasonable maximum value for BMI indicate outliers are present; massive skew histogram	Investigate if outliers are due to incorrect values for HEIGHT; if so, do nothing; if not, remove	Removal of HEIGHT outliers will address the BMI outlier problem, because BMI is derived from HEIGHT
SYS.B.P	Has a maximum value that is not reasonable for systolic blood pressure, and that is much further beyond the 3rd quartile than the 3rd quartile is beyond the median	Replace with a missing value, and add to data quality plan to impute a value	Only the one outlier, so imputation will result in no loss of data

(d) No sampling to be done. The small number of instances with missing values for the target feature will, according to the data quality plan, either me removed or imputed. The two class values, true and false, have very equal frequencies, so classes are not imbalanced.

Question 4

(a) The histogram for the different levels of the TACHYCARDIA feature are slightly different suggesting that there is a relationship between the DIA.B.P. and TACHYCARDIA features.

(b) It looks like patients with higher diastolic blood pressure values are more likely to suffer with tachycardia than those with lower blood pressure values.

(c) True. The histograms for the the HEIGHT feature split according to the different target levels look very similar. This suggests that there is no strong relationship between the HEIGHT feature and the target feature, TACHYCARDIA .

(d) No, the missing values do not present a problem. Although there are many missing values for the PREV. T ACHY. feature, they are equally prevalent for both values of the target feature. This is illustrated by the even split between the target levels in the bar on the left hand side of the bottom chart.

(e) When a value for PREV. TACHY. is present it is quite predictive of the target feature: previous occurrences of tachycardia are predictive of feature occurrences.